The effect of coups on military expenditure

Econometrics III (Course No. 4352)

[redacted]

2026-07-02

Table of contents

1 Motivation **3**

2 Literature **4**

3 Method **6**
3.1 Variables . . . 6
3.2 Threats . . . 7
3.2.1 Measurement Errors . . . 7
3.2.2 Non-Ignorability and OVB . . . 8
3.2.3 Simultaneity and time frames . . . 9
3.3 Identification strategy . . . 9
3.3.1 Addressing Measurement Errors . . . 9
3.3.2 Control Variables . . . 10
3.3.3 Differences-in-Differences . . . 10
3.4 Data and Data requirements . . . 12
3.4.1 Data military expenditures . . . 12
3.4.2 Data Coup . . . 12
3.4.3 Data Regimes . . . 13
3.4.4 Other Controls . . . 13
3.5 Summary Statistics . . . 14
3.5.1 Panel Data . . . 14
3.5.2 Successful and Unsuccessful Coup Countries . . . 14

4 First Results **18**
4.1 Control group . . . 18
4.2 Two-way Fixed Effects model . . . 18
4.3 Callaway and Sant Anna (2021) . . . 20
4.3.1 Prerequisites . . . 20

5 Conclusion and outlook **23**

6 References **24**
6.1 Literature 24
6.2 Data 24

1 Motivation

While thinking about possible research questions and projects we encountered many difficulties especially when we started to think about feasible modelling methods, data availability and the fact that a lot of the research questions would rely on time series data. That is why we decided to just focus on a topic that we are generally interested. We are intrigued by history, politics and obviously economics so we decided on a topic that is combining these interests. The broader topic we are aiming to work on is the connection between the form of government and military spending. More particularly, we would like to examine if dictatorships or autocratic forms of government have higher military spendings per capita.

Since the invasion of Ukraine by Russia in 2022, military spending increased all over the world. It has become a prominent topic and will shape the political and economic discussions for the next years. Germany, for instance, has nearly doubled their military expenditures from 2010 to 2025 (+83%). Therefore, it is of high interest to understand the key drivers of military spending. Then, we asked ourselves if there could be a link between military expenditures and the form of government.

As there already exists literature on a possible connection of military spending and the form of government without absolute results we thought of narrowing down my research question. We decided to focus on sudden changes of government induced by force-generally referred to as coup. This also opens opportunities with respect to possible modelling as we thought of a similar approach to what was discussed in class about the research on the influence of gang violence on crime. It would be beneficial to be able to compare some before and after situation in combination with a treatment. The treatment in this case could be the exactly the installing of a regime via a coup d'état e.g. of the military. So, we plan to consider the coup a random treatment that some countries get and some not. The final research question then would be **if there is an effect of both the forceful installment of a regime as well as the unsuccessful attempt to install one on the share of military expenditures on government spending**. The causal claim would be that there is a positive effect on

spendings as the military or new holder of power probably would like to strengthen the regime while there frequently is an inclination to use force to achieve this. There are of course a lot of other factors that influence military spendings we will try to account for in the following project. Our work aims to clarify if there are indicators that show the connection between a coup and military spending. It would thereby contribute a small part to the broader topic of the behavior of authoritarian regimes. Although this project does not explicitly aim at these kind of regimes a coup generally leads to a form of extractive government. Besides that, it adds to the research about government spending priorities in times of transition.

2 Literature

First we have to define what a coup is. Then we will discuss the existing literature. Coups are defined as successful "illegal and overt attempts by the military or other elites within the state apparatus to unseat the sitting executive" that leads to a new government where the perpetrators "seize and hold power for at least seven days" (Powell and Thyne 2011, 252).

There is extensive literature on the connection between authoritarian regimes vs democracies and their military expenditures and not all authors come to the same conclusion.

Fordham and Walker (2005) discussed if there are historical patterns supporting the claim that democracies tend to have lower military expenditures. They argue that the institutional structure of democracies usually leads to lower levels of military expenditures since politicians have to justify their decisions in order to get reelected. Furthermore, they assume that people generally prefer investments in public goods (such as education or health care) rather than military expenditures because the general public can directly benefit from the public goods. Additionally, autocratic regimes do not have to justify their government spending decisions in order to get reelected.

Goldsmith (2003) argues that military expenditures in autocratic regimes are higher as it is used to suppress internal threats.

Regarding coups Belkin and Schofer (2003) claim that high military expenditures are used to secure the loyalty of the military in autocratic regimes. The authors also show that military expenditures stay higher than economically reasonable after a coup. Hence, the high level of military spending slows economic development down.

In contrast, Collier and Hoeffler (2007) find that high military expenditures increase the risk of a coup as the military becomes more powerful.

Overall, not all authors come to the same conclusion. Some argue that military expenditures are also influenced by neighboring countries and the general level of conflict in the region. It is plausible that governments adjust their expenditures in response to a successful or unsuccessful coup in a neighboring country. This would need a more complex model to account for spatial auto-correlation.

Dunne, Perlo-Freeman, and Smith (2009) take a closer look at spill over effects in panel data. The authors show that even between two mutually hostile powers, the military expenditures may depend on the relationship between the two countries and other occurring events in the region. They conclude that military decisions are not made in isolation and hence, errors are not independent across units. Therefore, they suggest to use spatial error models or spatial autoregressive (SAR) models as a more appropriate methodological approach.

While current literature has mainly studied the determinants of military spending and driving factors of coup risk, not much attention has been paid to the reverse relationship. How do military budgets evolve after a coup has occurred? It remains unclear whether post-coup governments increase military spending to consolidate power and reward the loyal military, or whether spending decreases due to economic instability or efforts to signal a return to civilian governance.

Our project aims to expand the scope of the literature by trying to explain what happens to military budgets when coups occur.

3 Method

To answer the question if the forceful installation of a regime increases military expenditures, we first have to find an appropriate model and define the respective variables. We decided to construct a linear regression model following a Difference in Differences approach for identification. Firstly, we will describe the variables of interest and explain why they are important for my analysis. Afterwards, we will focus on potential threats for identification and provide solutions in the form of an identification strategy.

3.1 Variables

The military expenditure/government spending ratio lnMILDGS, describing the share of military expenditures of government spending, is our dependent variable. We decided to work with a ratio as it accounts for differences in income between countries. This is necessary as countries with a higher GDP on the one hand and relatively rich countries (GDP/capita) on the other tend to have higher total military expenditures. Logarithms are useful in this case since they ensure appropriate interpretability and moreover account for possible skewness issues. Interpretability is enhanced because variation can be regarded as percentage changes. Right-skewness is a major problem in income distributions. With respect to our variable of interest there is also the risk of right-skewness due to the fact that there are countries with exceptionally high shares of military expenditures.

Coup will act as a dummy variable. SuccessfulCoup equals 1 in all years after a successful coup has occurred, and 0 otherwise. As many countries will have experienced a coup more than once the explanatory power is limited. Unsuccessful coup follows a similar logic to the successful coup following the same definition but inherently without success.

Regime type is essential for our attempt to identify the effect of a coup as 1) coups tend to occur more frequently in combination with specific regime types and 2) we need to account for different shares of expenditures in different regimes. We will differentiate between ten

regime types 0 Parliamentarism, 1 Semi-presidentialism, 2 Presidentialism,3 Semi-monarchy, 4 Party-based rule, 5 Personalist rule, 6 Military rule, 7 Absolute monarchy, 8 Oligarchy, 99 Occupation, civil war or otherwise unclear. (Political Regimes of the World Dataset, v.3.0., 2025)

Inequality is potentially influencing both the military expenditures of a country and the probability that a coup occurs. Political and social unrest is to a certain extent caused by inequality. In order to tackle agitations within a country armed forces are frequently used. However, if inequality becomes unbearable for society coups are a common phenomenon to enforce a change of government. Hence, inequality acts as confounder and we have to control for it. In the model, inequality will be measured by the GINI coefficient.

It is of utmost importance to account for periods of war since they are a major natural driver of military expenditures. War would be incorporated into the model as a binary variable that takes on value 1 for each year of war and 0 for a year in peace. Again, one could argue that war is also influencing the probability of a coup and therefore we have to control for confounder.

lnAF describes the logarithm of the armed forces per 1000 people. The logarithm is again used for skewness reasons. We are once more addressing a confounder issue since the relative size of armed forces strongly correlates with military expenditures and explains a lot of variation in this context.

The logarithmed growth rate of GDP, GDPGrowth, influences social welfare capabilities, inequality, military capacities and many more factors which are influencing either the share of military expenditures and other already integrated model variables.

3.2 Threats

3.2.1 Measurement Errors

One problem that our data probably suffers from are measurement errors. While liberal democracies experience internal pressures to provide correct data about government spending,

authoritarian regimes are prone to misreporting numbers. There are several reasons why this could be the case. Depending on what a regime wants to achieve, military expenditures could be overstated to assert strength or underreported to reassure potential enemies. Whatever the reasons are the consequence would be an attenuation bias. In the context of our analysis underreported military expenses after a coup in order to calm potential rivals could introduce bias by reducing the coefficient of the coup treatment. Measurement errors could also occur with independent variables. For instance, the borders between specific regimes are to a certain degree fluent which could lead to misclassifications that distort the results.

3.2.2 Non-Ignorability and OVB

One of the main threats to causal identification is also that the treatment "coup" is not ignorable. Coups do not happen randomly most of the time but are rather a product of the economic, political and social situation in a country. The regional security setting, global conflicts or economic crisis and inequality within a country are examples for this. However, if we are able to control for the most important confounding effects there are still random elements that make a coup occur and succeed which makes identification possible. For this reason it would be beneficial to be able to compare very similar initial situations where the only difference is the success of a coup. The problem of non-ignorability is implying another potential threat: the omitted variable bias. As already touched upon in the "Variables" segment there are several factors we have to control for since they are influencing both the probability of the occurrence of a coup and the respective variable like inequality. As we face panel data we furthermore have to deal with the problems of country-and time-specific effects. There are time-invariant characteristics of a country that affect military spending as well as time-dependent global events such as the 2008 financial crisis we have to account for.

3.2.3 Simultaneity and time frames

One could also argue that there are simultaneity issues between the independent variables as well as between the dependent and independent variables. With respect to the dependent and independent variables the effect of military spending and coups is hard to disentangle. Higher military expenditures strengthen military capacity and thereby the probability of a coup. Simultaneously higher expenses are used to consolidate power or suppress agitations against the government reflecting the high probability of a coup. That means they are both influencing each other at the same time. Regarding dependent variables Coups a particularly often started by the army but coups are also happening often within a military regime. One could make a similar point for army spending and the probability of war. Nevertheless, there is a lot of room for interpretation. In some cases higher military spending could decrease the probability of war due to a new power asymmetry. Lastly, there are difficulties in choosing correct time frames to measure the impact of the coup correctly. It is unrealistic that there are immediate effects on the budget but rather happening a few years after a successful coup. That means some of the coups acknowledged successful by the database will not exist at the time of the measurement which also poses a threat.

3.3 Identification strategy

3.3.1 Addressing Measurement Errors

When it comes to the potential misreporting of data it is important to rely on independent and credible data portals that do not just accept government figures but base their publications on their own research. In the case of military expenditures SIPRI is an example for an institute like this. We will further discuss these aspects in the data section of this project. Other remedies for measurement errors are using logarithms since they minimize the impact of outliers due to misreporting as well as using different databases for the regime classification for example. If independent portals provide similar assessments of a regime type robustness is strengthened.

3.3.2 Control Variables

In this section we will refer to potential threats and point out how we could best causally identify coups as a reason for rising military expenditures in military regimes. The first problem we are trying to tackle is the non-ignorability. Best practice in this case would be to carry out experiments where we randomly assign a treatment. That is of course not possible in the context of coups and countries. Consequently, we have to control for confounders as well as country and time specific effects in order to transform coup into a treatment that comes as close as possible to a quasi-exogenous variable. Controlling for confounders like inequality or GDP growth ensures that changes in expenditure are not falsely assigned to coups but to their real causes. It is neither good for the model nor possible to cover all confounders, but one has to identify the most important ones in order to confine the resulting OVB. By including a time and country specific component we account for global trends in military spending, conflictridden and rather peaceful periods as well as cultural and geographical circumstances. This is crucial as we want to conduct within-country comparisons that should not be influenced by individual or time specific factors.

3.3.3 Differences-in-Differences

The method that addresses the necessities for causal identification for this problem set most appropriately is a Differences-in-Differences (DiD) approach. The idea is to balance the confounders mentioned above across a treatment group and a control group in order to create very similar groups of countries that would have followed a common trend without the treatment (coup). We can still include controls and fixed effects to further improve precision. As standard DiD would require a treatment to be assigned at the same point in time for all the treated subjects we have to find an alternative since coups do not fulfill these prerequisites. A solution for that is Staggered DiD. Staggered DiD creates flexible comparison groups by only referring to the subjects of the control group at a specific point in time that have not yet experienced a coup. To furthermore strengthen the causal argument, we will include a

second treatment-unsuccessful coup. Countries that face coup attempts regardless of whether the attempt is successful are probably exposed to similar circumstances like social unrest or stronger military influence and the simultaneity issues outlined above. By comparing successful coups to unsuccessful ones and both to a control group we add another layer and are able to make claims 1) about whether the environment of coup attempts has an influence on military expenditures and 2) the successful implementation has an additional effect. Furthermore, this approach allows to tackle simultaneity by ensuring that the regime transition is triggering the increased military spending as both countries that face a successful and unsuccessful coup are exposed to the same high tension environment. This approach leads to the following regression equation:

$$\ln(\text{MILDGS}_{it}) = \beta + \beta_1 \cdot \text{SuccessfulCoup}_{it} + \beta_2 \cdot \text{UnsuccessfulCoup}_{it} + X'_{it}\gamma + \mu_i + \lambda_t + \epsilon_{it}$$

- β … Intercept (includes control group countries)
- X'_{it} … vector of control variables
- μ_i … country-specific fixed effects
- λ_t … time-specific fixed effects
- ϵ_{it} … error term

While standard DiD would contain a treatment-control group difference, a time effect and an interaction term between treatment and time effect staggered DiD uses explicit country and time specific effects. As a consequence the coefficients for Successful Coup and Unsuccessful Coup capture the interaction term by representing a country after a successful or unsuccessful coup respectively.

3.4 Data and Data requirements

The research question requires a panel dataset observing countries over time. With standard linear regression it would be beneficial to include as many observations as available. However, our research design follows a DiD-approach which means that we have to build groups of countries with matching characteristics. In practice that means we are creating a group of countries that experienced a successful coup, one group with an unsuccessful coup and a control group with no coups. Countries across the group should have matching characteristics in key areas, mainly the already identified confounders, in order to make comparisons useful. That is because DiD requires parallel trends before coups as we want to distill an intervention effect. As there are no perfect matching countries, we would further include important control variables and fixed effects.

3.4.1 Data military expenditures

To start we need a reliable data source for our dependent variable, the share of military expenditures. The Stockholm International Peace Research Institute (SIPRI) is an independent and renowned institute focusing on disarmament, peace and military expenditures. It lists the shares of government spending dating back to 1988 for the vast majority of countries. Problems could arise as many countries of interest that are actually affected by coups have missing data which could lead to a selection bias in the results. Nevertheless, this is the most reliable data source available.

3.4.2 Data Coup

For the Successful and Unsuccessful coup countries variable we will make use of the Coups détat dataset by Jonathan Powell and Clayton Thyne. They are identifying successful coup events according to the definition provided in the "Variables" segment. Events are listed starting from 1950 until 2024 in the updated version. By matching the coups data to the

available shares of government spending the major part of the core mechanism of my model is covered.

3.4.3 Data Regimes

In order to capture the differences of regimes the "Political Regimes of the World Dataset, v.3.0." by Carsten Anckar and Cecilia Frederiksson lists types of regimes until 2025 and encapsulates the different dimensions of regimes the best. An alternative dataset, the "Political Regime Dataset" provided by the University of Texas Inequality Project (UTIP) is used in a reference paper. However, a major drawback is that they are only providing data up to 2006.

3.4.4 Other Controls

Turning to the other control variables the data used for the War variables stems from the Correlates of War Project. They differentiate between intra-state war and inter-state wars. Data entries are available until 2007. For our model we would merge the two variables into one War variable as it still fulfills its main purpose to capture confounding effects. For the inequality control the UTIP is providing an inequality dataset (EHII Dataset,2015) measuring inequality within countries by the GINI coefficient until 2015. There is a risk for a selection bias as countries with unstable and weak governments often lack valid statistics. For the GDP growth the World Bank issues reliable data up until 2024. For our model we will use the annual GDP growth rate in constant USD as we want to account for inflation and measure real GDP growth.

3.5 Summary Statistics

3.5.1 Panel Data

First, we created a panel data frame encapsulating some of the key variables: milexpenditures, succesful_coup, unsuccessful_coup and regime_type. The reason why we have not included other control variables yet is data availability as many entries would have to be discarded because of missing entries. It is also not necessary with a DiD approach. However, they still play an important role in the matching process even if they are not included in the final model at the end. While forming the panel data set one obstacle was that there are multiple successful coups or respectively unsuccessful coups for countries in a single year. To maintain unique year/country entries we had do discard these entries. This, however, has no effect on the results as we use years as time unit.

```
Unbalanced Panel: n = 200, T = 2-35, N = 6721
```

The panel includes 6721 observations from 1990 to 2025 and is unbalanced since not every country has the same number of entries. The time frame reaches from 2 observed years to 35 observed years with all in all 200 countries.

3.5.2 Successful and Unsuccessful Coup Countries

The next step is to get an overview concerning the two treatment groups. There are overall 59 entries for successful coups and 46 for unsuccessful coups. Since they are lacking data for military expenditures, we have to discard 30 data entries for successful coups and 9 for unsuccessful coups. The following countries have valid entries for military expenditures, the coup variables and regime data in the coup year:

Table 1: Overview Successful Coups

Country	Year	Milexpenditures
azerbaijan	1994	0.1327
azerbaijan	1995	0.1446
bangladesh	1996	0.1199
bolivia	2024	0.0367
burkina faso	2015	0.0614
burundi	1993	0.1383
burundi	2001	0.2390
burundi	2015	0.0920
chad	2006	0.2150
djibouti	2000	0.1436
ecuador	2000	0.0625
gabon	2019	0.0909
guinea-bissau	1998	0.0563
guinea-bissau	2000	0.1038
guinea-bissau	2010	0.0983
guinea-bissau	2022	0.0671
lesotho	2014	0.0377
madagascar	1992	0.0384
madagascar	2010	0.0464
mali	2012	0.0770
niger	2021	0.0559
nigeria	1990	0.0223
papua new guinea	1990	0.0588
papua new guinea	2012	0.0232
paraguay	2000	0.0635
peru	2000	0.0833
philippines	1990	0.1001
sudan	2021	0.0950
togo	1991	0.1148

The next steps are to check if these countries have valid entries for all the years observed. If not, there are usually two options: Either one discards all observations for the respective country or one adjusts the time frame if the lacking data points are at the edge of the time frame. Afterwards it is necessary to identify suitable control countries by ensuring balanced confounders across the groups in order to fullfil the parallel trend assumption of DiD. The fixed effects included in the model pose some relief regarding the rigor of having balanced confounders.

Table 2: Overview Unsuccessful Coups

Country	Year	Milexpenditures
algeria	1992	0.0699
bangladesh	2024	0.0690
burkina faso	2014	0.0608
burkina faso	2022	0.0919
burundi	1996	0.2060
cambodia	1997	0.2419
chad	2021	0.1495
egypt	2011	0.0601
egypt	2013	0.0465
fiji	2000	0.0707
fiji	2006	0.0648
gabon	2023	0.0596
guinea	2021	0.0996
guinea-bissau	2003	0.0798
guinea-bissau	2012	0.1799
honduras	2009	0.0379
lesotho	1991	0.0931
lesotho	1994	0.0688
madagascar	2009	0.0590
mali	2020	0.1302
mali	2021	0.1257
mauritania	2005	0.1065
mauritania	2008	0.1195
myanmar	2021	0.1744
niger	1996	0.0667
niger	1999	0.0622
niger	2010	0.0598
niger	2023	0.1250
nigeria	1993	0.0180
pakistan	1999	0.2398
rwanda	1994	0.2144
sudan	2019	0.0880
thailand	2006	0.0516
thailand	2014	0.0622
togo	2005	0.0815
ukraine	2014	0.0663
zimbabwe	2017	0.0681

4 First Results

4.1 Control group

As the successful and unsuccessful coup entries primarily cover South-American, South-EastAsian , South-Asian and African countries we will focus on these areas for my control group. They roughly match regarding inequality, GDP growth and face rather similar conflicts as multiple countries of the same region are affected at the same time from wars thereby balancing confounders. Moreover, the number of armed forces does usually match quite smoothly for similar reasons. Besides that, regime type and fixed effects account for a significant part of variation. Hence, we revisited the SIPRI dataset and discarded observations from countries that are not in the regions mentioned above.

4.2 Two-way Fixed Effects model

For this model a two-ways within panel linear model is used. It introduces bias as it automatically gets rid of years with no observations. By reducing the sample to 72 countries and 2035 observations there, it is still an unbalanced panel. However, the time span for a country is now at least 11 which is ameliorating fixed effects and strengthens the claim that we are observing pre-and post-treatment environments.

```
Twoways effects Within Model

Call:
plm(formula = milexpenditures ~ successful_coup + unsuccessful_coup +
    regime_type, data = panel_final, effect = "twoways", model = "within",
    index = c("country", "year"))

Unbalanced Panel: n = 72, T = 11-35, N = 2035


Residuals:
    Min.  1st Qu.   Median  3rd Qu.     Max.
-2.62171 -0.16393 -0.00458  0.16388  1.20347

Coefficients:
                  Estimate Std. Error t-value Pr(>|t|)
successful_coup   -0.04326    0.06715   -0.64    0.520
unsuccessful_coup  0.11939    0.06027    1.98    0.048 *
regime_type        0.00249    0.00106    2.34    0.019 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    198
Residual Sum of Squares: 197
R-Squared:      0.00557
Adj. R-Squared: -0.0502
F-statistic: 3.59911 on 3 and 1926 DF, p-value: 0.013
```

First results of the reduced model point to non-significant effects of a successful coup. Success-

ful coup although non-significant would even have a negative impact. However, we can find a significant positive effect of the share of military expenditures of government spending after an unsuccessful coup. As we used the natural logarithm of military expenditures, we can say that an unsuccessful coup increases the share of military expenditures of government spending by approximately 12% which could point to a reaction of current holders of power. The idea would be that after an unsuccessful coup, political leaders are alarmed and increase military spending to secure their power. Overall, the selected variables are significant together based on the F-statistic with very low explanatory power (R-squared).

4.3 Callaway and Sant Anna (2021)

There are also alternatives to the modelling approach for Staggered DiD that we used. A main problem is that the current model measures rather a constant effect after a coup. That means if a coup happened in the 90s, the model calculates an average effect of the remaining data points. Furthermore, early and late treated countries are compared with each other which can introduce bias when the effect changes over time. Another point of critique is that staggered DiD would imply persistent treatment which is not necessary the case for coups: There can occur a form of "double-treatment". A possible solution would be estimating group-time effects as proposed by Callaway and Sant Anna (2021). It would enhance the measurement of the direct effects of a coup. However, this approach poses a new threat as it is less tolerant towards lacking data which could lead to a very small number of observations.

4.3.1 Prerequisites

Callaway and Sant Anna´s approach requires that there is no double treatment. As a consequence, only the first successful or unsuccessful coup is considered in the analysis. Furthermore, we make use of 2-year sampling in order to increase the treatment groups. If there was only one treated observation within a year we would have to drop it in the Callaway Sant'Anna framework since we could not calculate group time average as there is no variation in the treated

group. This leads to a sample size of 14 for successful coups and 15 for unsuccessful coups. We again use the logarithm of military spending. The results of ATT´s by Callaway and Sant Anna can be interpreted as average percentage change in military expenditures/Government spending for a treated country for a specific year after treatment compared to a clean control group independent of calendar-time trends.

Callaway and Sant Anna depict their results in an event study plot. The x-axis represents the years before and after the coup event and the y-axis the effect size on the logarithm of military expenditures. The plot is bounded at -6 to +6 years in order to focus on the effects of a coup. The results show a significant increase in military expenditures after an unsuccessful coup while there is no significant effect for successful coups. The parallel trend assumption for unsuccessful coups represented by a pre-treatment effects size of 0 accurately for unsuccessful coups and reasonably for successful coups. These results also hint to a stepwise increase of military expenditures after an unsuccessful coups to secure power. For successful coups it could be argued as you have a possible late effect that the new regime aims to cement power via military spending after a first consolidation phase. However, it is important to note that the number of observations is very low which makes it hard to draw strong conclusions from these results. Moreover, the confidence intervals are quite wide which also points to a high level of uncertainty.

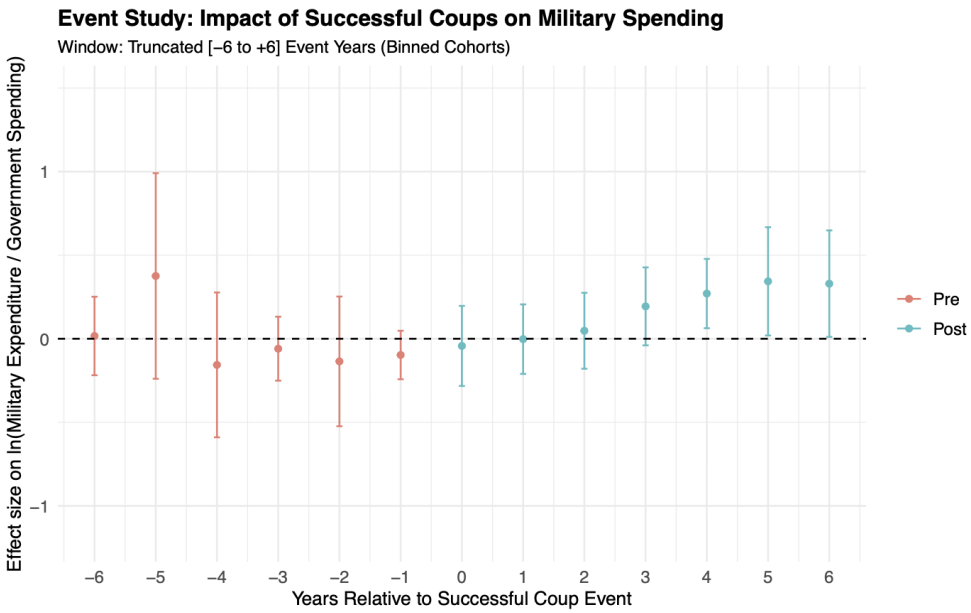


Figure 1: Event Study Plot for Successful and Unsuccessful Coups

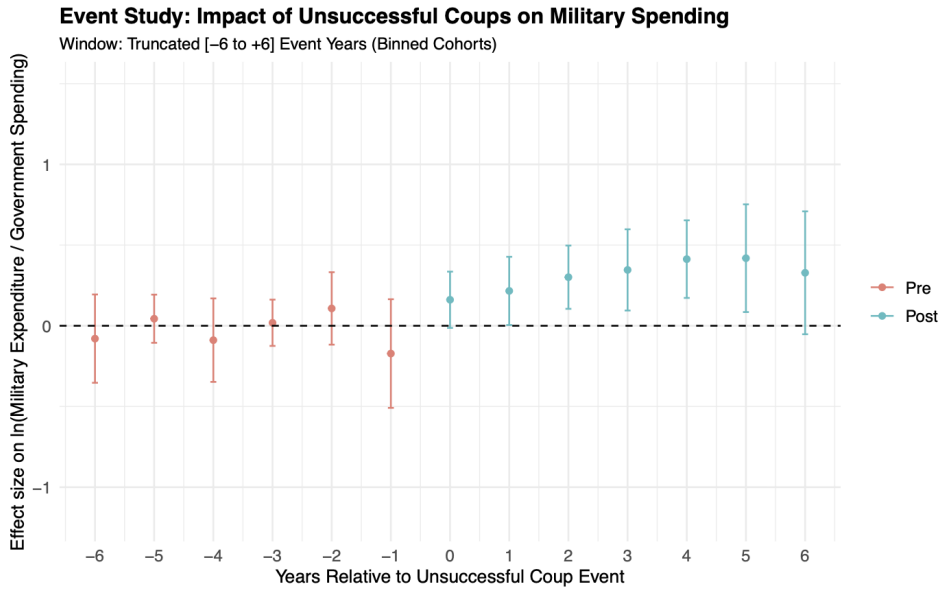


Figure 2: Event Study Plot for Successful and Unsuccessful Coups

5 Conclusion and outlook

In spite of a methodological framework that should be able to capture the causal impact of a coup on military expenditures to a certain extent the validity of these first result is limited. A major problem is a lack of consistent data and imprecise or wrong reporting of countries that are affected. One could continue the work by including more controls or narrowing down the scope of the analysis. There are also alternatives to the modelling approach for Staggered DiD that we used. A main problem is the current model measures rather a constant effect after a coup than the exact effect of the coup. That means if a coup happened in the 90s, the model calculates an average effect of the remaining data points. Considering that there is no specific point in time where a coup displays its influence it is still a reasonable approach. Another point of critique is that staggered DiD would imply persistent treatment which is not necessary the case for coups: There can occur a form of "double-treatment". This problem can be mitigated by estimating group-time effects as proposed by Callway and Sant Anna (2021). We implemented it because it enhances the measurement of the direct effects of a coup. However, this approach poses a new threat as it is less tolerant towards lacking data which leads to a very small number of observations as discussed in the previous section. For a proper thesis it would be beneficial to focus on a small region in a specific time period instead of making a general causal claim of coups on military expenditures. A possible idea would be to look at Arabic countries after the Arabic Spring. Although this would threaten external validity it would probably provide a more accurate picture of the effect in a specific context.

6 References

6.1 Literature

Belkin, A. & Schofer, E. (2003): Toward a Structural Understanding of Coup Risk. Journal of Conflict Resolution, 47(5),p. 594–620.

Callaway, B., & Sant'Anna, P. H. (2021). Difference-in-Differences with multiple time periods. Journal of Econometrics, 225(2), p. 200-230 Paul Collier & Anke Hoeffler, 2006. "Military expenditure in post-conflict societies," Economics of Governance, Springer, vol. 7(1), pages 89-107.

Dunne, J. P., Perlo-Freeman, S., & Smith, R. P. (2009). Determining military expenditures: Arms races and spill-over effects in cross-section and panel data (Working Paper No. 0901). Bristol Business School, University of the West of England.

Fordham, B. O., & Walker, T. C. (2005). Kantian Liberalism, Regime Type, and Military Resource Allocation: Do Democracies Spend Less? International Studies Quarterly, 49(1), 141–157.

Goldsmith, B. E. (2003): Bearing the Burden: A Cross-National Analysis of Military Spending. Journal of Conflict Resolution, 47(5), p. 551–573.

Powell, J. M., & Thyne, C. L. (2011). Global Instances of Coups from 1950 to 2010: A New Dataset. Journal of Peace Research, 48(2), p. 249–259

6.2 Data

Anckar, C. (Creator), Fredriksson, C. (Creator), Anckar, C. (Contributor) (15 May 2025). Political Regimes of the World Dataset, v.3.0. Harvard Dataverse. 10.7910/dvn/ak8nvx

Correlates of War Project: Correlates of War Project (o.J.). War Datasets (v4.0). Available under: UTIP (University of Texas Inequality Project - Regime Data):

University of Texas Inequality Project (UTIP) (o.J.). Political Regime Dataset. Austin: University of Texas. https://utip.lbj.utexas.edu/. https://correlatesofwar.org/ (Accessed on:

[29 Jan 2026).

UTIP (EHII Dataset): Galbraith, J. K., et al. (2015). Estimated Household Income Inequality (EHII) Dataset. University of Texas Inequality Project

Dunne, J. Paul, Sam Perlo-Freeman, and Ron P. Smith. 2009. "Determining Military Expenditures: Arms Races and Spill-over Effects in Cross-Section and Panel Data." Working Paper 0901. Bristol Business School, University of the West of England.

Powell, Jonathan M., and Clayton L. Thyne. 2011. "Global Instances of Coups from 1950 to 2010: A New Dataset." *Journal of Peace Research* 48 (2): 249–59. https://doi.org/10.1177/0022343310397436.